

IntentTalk: Future-Grounded Motion Planning for Event-Aware Co-Speech Motion Generation

– Supplementary Material –

A. Datasets

We conduct experiments on two co-speech motion datasets, BEAT2 and GES-X. BEAT2 is used for model training and in-domain evaluation, while GES-X is used exclusively for zero-shot cross-dataset evaluation. This protocol allows us to evaluate both the generation performance under the training-domain distribution and the generalization ability of the proposed future-planning mechanism under unseen speakers and motion distributions.

1) *BEAT2*. BEAT2 [1] is a large-scale multimodal conversational motion dataset designed for holistic co-speech motion generation. It provides temporally aligned speech, linguistic information, and high-quality 3D human motion represented with SMPL-X, covering coordinated facial expressions, body motion, hand articulation, and global movement. Compared with earlier upper-body gesture datasets, BEAT2 supports the evaluation of holistic motion generation and cross-part coordination.

Following the official evaluation protocol, we use the English subset and split the data into training, validation, and test sets with a ratio of 85%:7.5%:7.5%. Motion sequences are sampled at 30 FPS and represented using 330-dimensional 6D SMPL-X joint rotations together with 3D root translations, four contact labels, and 100-dimensional FLAME-compatible facial expression coefficients. Audio is sampled at 16 kHz. Each training clip contains 64 motion frames with a stride of 20, and the first four frames are used as seed motion.

We report results under both the *one-speaker* and *all-speaker* settings. The former evaluates generation under a more controlled speaker-specific distribution, whereas the latter provides a more challenging setting involving greater inter-speaker variation in gesture style and motion dynamics. BEAT2 is also used for all ablation, planning, and representation analyses unless otherwise specified.

2) *GES-X*. GES-X [2] is a large-scale in-the-wild co-speech gesture dataset containing diverse speakers, speech content, motion patterns, and recording conditions. Compared with the relatively controlled distribution of BEAT2, GES-X exhibits substantially larger variations in speaker identity, gesture style, motion amplitude, and capture environment, making it suitable for evaluating cross-dataset robustness.

In our experiments, GES-X is used only for *zero-shot evaluation*. Neither the future motion-plan tokenizer nor the speech-to-plan-to-motion generator is trained or fine-tuned on GES-X. The model trained on BEAT2 is directly applied to GES-X using the same inference pipeline and evaluation protocol.

Therefore, the resulting performance reflects the transferability of the learned motion representation and future-planning mechanism rather than adaptation to the target dataset.

This cross-dataset setting is particularly relevant to IntentTalk because the learned future plans are expected to capture recurring structural properties of upcoming motion rather than speaker- or dataset-specific motion instances. Consistent performance on GES-X therefore provides complementary evidence for the generalization of the proposed planning representation under distribution shift.

B. Additional Implementation Details

All experiments are conducted on a single NVIDIA A100 GPU. Following the protocol of EMAGE [1], motion sequences are sampled at 30 FPS and represented by 330-dimensional 6D SMPL-X joint rotations, 3D root translations, four foot-contact labels, and 100-dimensional FLAME-compatible facial expression coefficients. Audio is sampled at 16 kHz. For speech conditioning, we use onset and amplitude features, 1,024-dimensional HuBERT representations, 300-dimensional fastText word embeddings, and CLIP ViT-B/32 sentence embeddings.

Each training sample contains 64 motion frames with a stride of 20, where the first four frames are used as seed motion. The body backbone has a hidden dimension of 768, and each regional RVQ motion latent has 256 dimensions. The speech-to-plan predictor consists of a two-layer Transformer encoder with a feature dimension of 128, four attention heads, a feed-forward dimension of 512, and a dropout rate of 0.1.

The future motion-plan tokenizer contains two residual MLP blocks and a codebook with $K = 64$ entries of dimension $D_p = 128$. We use a future horizon of $H = 2$ motion tokens and an offset of $o = 1$, corresponding to the subsequent eight motion frames (approximately 0.267 s at 30 FPS). The VQ commitment weight is set to 0.25.

Training is performed in two stages. In Stage I, the future motion-plan tokenizer is pretrained for 100 epochs using Adam with $\beta_1 = 0.5$ and $\beta_2 = 0.999$, a batch size of 128, and a learning rate of 2×10^{-4} . After pretraining, the future-window encoder, decoder, and plan codebook are frozen. In Stage II, the speech-to-plan predictor, event-aware semantic gate, hierarchical plan-conditioned modules, and motion generator are jointly trained for 200 epochs with a batch size of 64 and an effective learning rate of 5×10^{-5} . We use no weight decay and clip the gradient norm to 0.99.

During training, discrete plan selection is implemented using straight-through Gumbel-Softmax with a temperature of

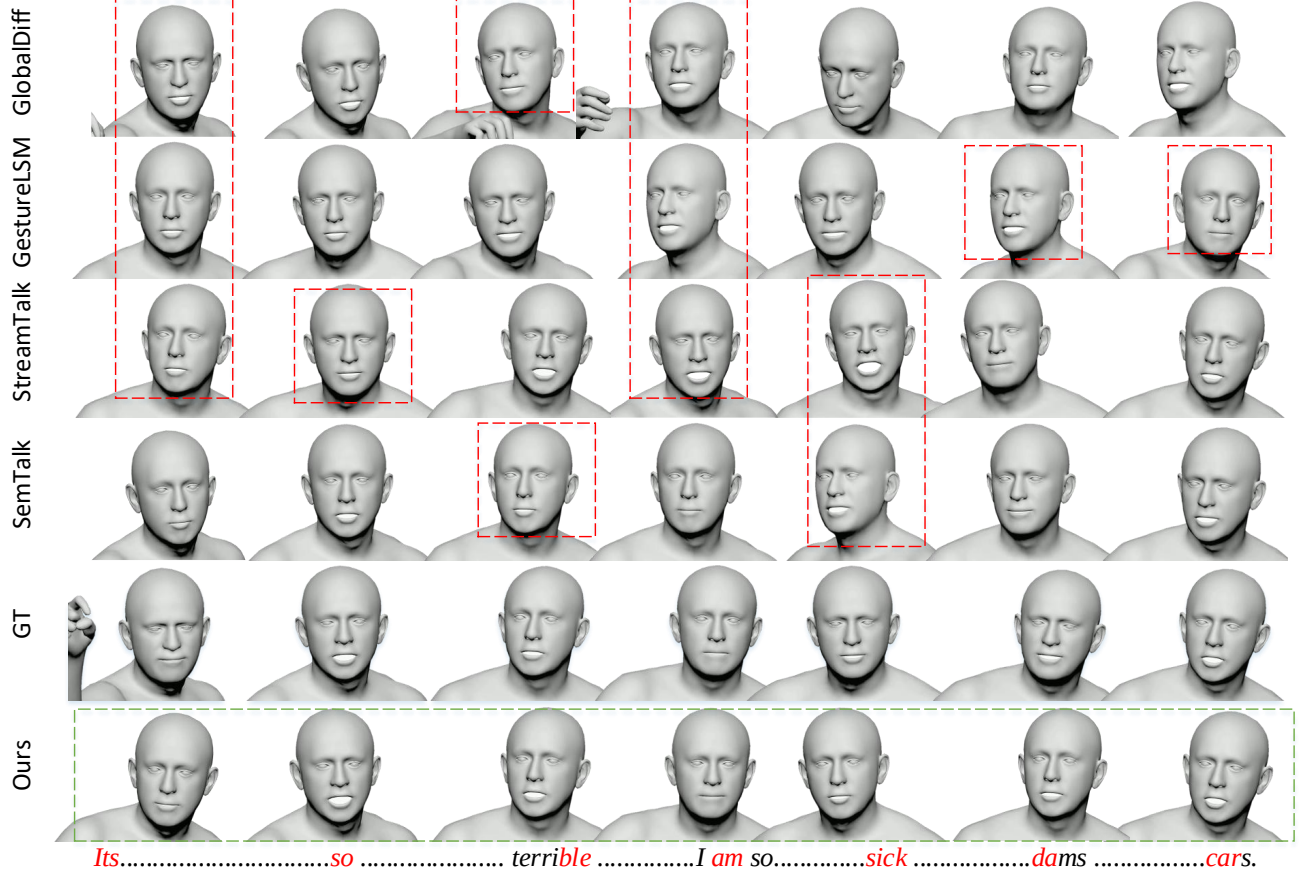


Fig. 1. Qualitative comparison of facial motion generation. Red dashed boxes highlight representative inconsistencies in competing methods. IntentTalk produces smoother and more temporally coherent facial dynamics across the speech sequence.

1.0. At inference, the plan index with the maximum prediction logit is selected deterministically. To stabilize optimization, the plan-related objective is disabled during the first 10 epochs and linearly introduced over the following 10 epochs.

We set $\lambda_p = 0.2$, $\lambda_{\text{align}} = 0.1$, $\lambda_{\text{ctr}} = 0.1$, and $\lambda_{\text{event}} = 0.5$. The contrastive temperature is set to 0.1, and all experiments use a random seed of 43 unless otherwise specified.

C. Qualitative Analysis

1) *Qualitative Comparison of Facial Motion*: Fig. 1 further compares the facial motion generated by different methods on the same speech segment. Although existing methods can produce plausible facial expressions at individual frames, they often exhibit noticeable temporal inconsistencies, as highlighted by the red dashed boxes. In particular, GlobalDiff, GestureLSM, StreamTalk, and SemTalk show occasional abrupt changes in head orientation, facial expression, or mouth configuration across neighboring frames, resulting in less coherent facial dynamics. In contrast, our method produces smoother transitions throughout the sequence while maintaining stable facial structure and more consistent speech-related mouth movements. The generated facial dynamics also exhibit a temporal evolution that is qualitatively more

consistent with the ground-truth sequence, suggesting that the proposed framework improves not only holistic body-motion organization but also the temporal coherence of facial motion.

2) *Qualitative Ablation Analysis*: As shown in Fig. 2, removing the event-aware semantic gate leads to delayed motion activation around the salient speech event, while discarding the future plan weakens the progressive organization from preparation to gesture realization. Without plan-conditioned FiLM, the motion becomes more localized, indicating reduced propagation of the planned motion across body regions. In comparison, the complete IntentTalk exhibits clearer temporal organization and more coherent motion realization. These visual patterns are consistent with the complementary roles of the proposed components: the gate regulates *when* motion is activated, the future plan provides the coarse evolution of upcoming motion, and hierarchical modulation governs *how* the plan is realized into detailed body motion.

D. Understanding the Learned Future Plans

To determine whether the learned discrete plans encode structured future-motion information rather than merely compressed motion features, we conduct linear probing on frozen representations. We compare the current motion token, the

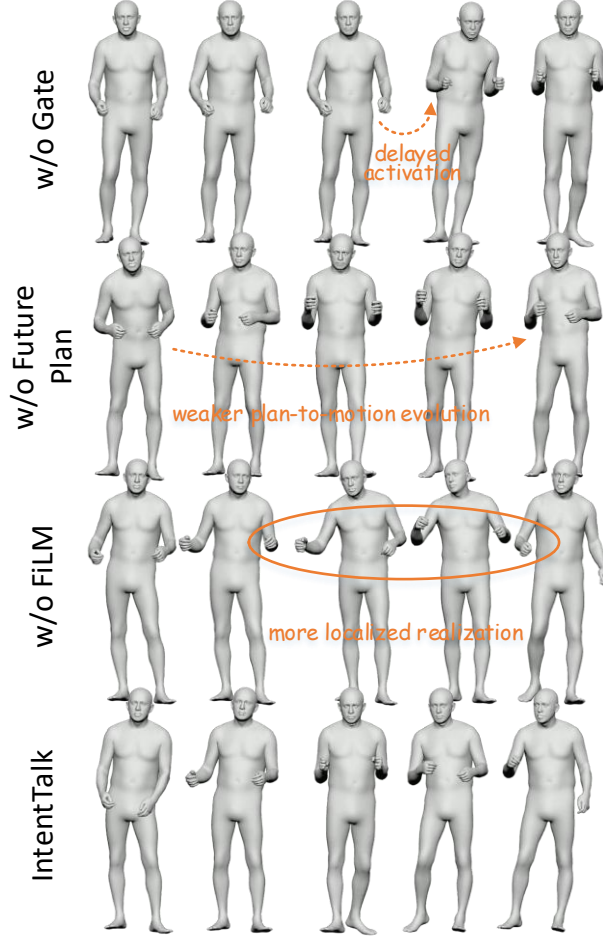


Fig. 2. Qualitative ablation comparison of IntentTalk. Removing the event-aware gate delays semantic-motion activation, removing the future plan weakens plan-to-motion evolution, and removing hierarchical FiLM results in more localized motion realization.

continuous future-window feature h_t^m , and the discrete future-plan embedding p_t^m . Four motion-derived attributes are considered: current body-part activation, future body-part activation, future motion energy, and future temporal evolution. The first serves as a negative control for temporal orientation, while the latter three characterize which body regions will participate, how strongly they will move, and how the motion evolves within the upcoming window. Classification tasks are evaluated using macro-F1 and motion-energy prediction using R^2 .

As shown in Table I, the current motion token achieves the highest performance on current body-part activation, confirming its strong representation of the instantaneous motion state. However, its predictability decreases substantially for future-oriented attributes. In contrast, both future representations provide stronger cues for upcoming body-part participation, motion magnitude, and temporal evolution. Although vector quantization introduces a moderate information loss compared with the continuous future feature, the discrete future plan retains most of its predictive capability, particularly for future body-part activation and motion evolution. This complemen-

TABLE I
LINEAR PROBING OF CURRENT AND FUTURE MOTION REPRESENTATIONS. ALL REPRESENTATIONS ARE FROZEN DURING PROBING. HIGHER VALUES INDICATE BETTER PREDICTABILITY OF THE CORRESPONDING MOTION ATTRIBUTES.

Representation	Cur. Part F1↑	Fut. Part F1↑	Fut. Energy R^2 ↑	Evolution F1↑
Current Motion Token	0.792	0.558	0.327	0.486
Continuous Future Feature	0.624	0.781	0.684	0.746
Discrete Future Plan	0.603	0.754	0.631	0.718

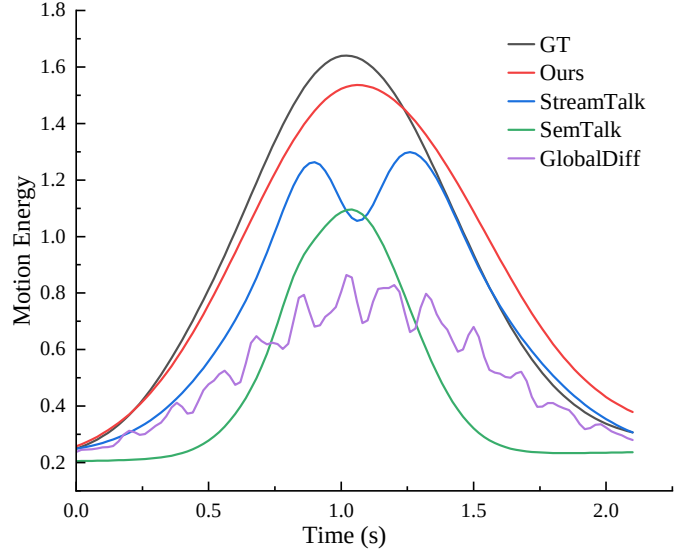


Fig. 3. Temporal evolution of motion energy for a representative speech segment. IntentTalk exhibits a smooth motion-energy profile with temporal evolution similar to the ground truth, while StreamTalk shows multiple local peaks, SemTalk produces a more localized response, and GlobalDiff exhibits frequent low-amplitude fluctuations.

tary pattern suggests that the proposed plan does not simply constitute a stronger generic motion feature, but preferentially encodes coarse structural properties of upcoming motion.

E. Temporal Motion Evolution

To characterize the temporal evolution of body motion, we compute the motion energy at frame t as the average frame-wise joint displacement:

$$E_t = \frac{1}{J} \sum_{j=1}^J \|\mathbf{x}_{t,j} - \mathbf{x}_{t-1,j}\|_2, \quad (1)$$

where J denotes the number of body joints and $\mathbf{x}_{t,j}$ is the 3D position of joint j at frame t . A larger E_t therefore indicates stronger instantaneous body motion, while the temporal profile of E_t reflects how motion intensity evolves over the speech segment.

Fig. 3 visualizes the temporal evolution of motion energy for a representative speech segment. The ground-truth motion exhibits a smooth temporal profile, gradually increasing toward a dominant motion event and then decreasing through a continuous transition. IntentTalk follows a similar global evolution pattern, with a comparable onset, peak location, and smooth

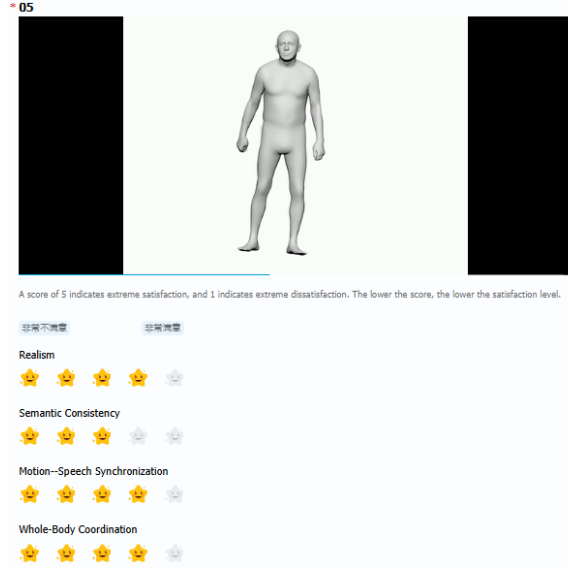


Fig. 4. Screenshot of user study website.

post-peak decay, although its peak magnitude is slightly lower than that of the ground truth. In contrast, StreamTalk captures the overall high-motion interval but exhibits two local peaks separated by an intermediate drop, indicating a less continuous temporal organization. SemTalk produces a more localized response with substantially lower motion intensity and a faster decay, suggesting that the generated motion is concentrated within a shorter temporal interval. GlobalDiff shows relatively low-amplitude and frequent local fluctuations throughout the sequence, resulting in a more fragmented motion-energy profile. These observations suggest that IntentTalk better preserves the coarse temporal evolution of the reference motion, supporting the effectiveness of future-window planning for organizing upcoming motion dynamics.

F. User Study Details

To complement the objective evaluation, we conduct a perceptual user study to assess the visual quality and speech-motion consistency of the generated motions. Following general subjective assessment principles established by the International Telecommunication Union (ITU) [3], we recruit 30 participants to evaluate four methods: SemTalk, GlobalDiff, StreamTalk, and IntentTalk. For each method, 10 test sequences are randomly selected, resulting in 40 evaluated videos and 300 individual ratings per method.

Participants evaluate each sequence using a five-point Mean Opinion Score (MOS) scale, where a higher score indicates better perceptual quality. Four complementary criteria are considered: *Realism*, measuring the overall naturalness and plausibility of the generated motion; *Semantic Consistency*, assessing whether the gestures are appropriate for the corresponding speech content; *Motion-Speech Synchronization*, evaluating the temporal alignment between speech and body

motion; and *Whole-Body Coordination*, measuring the coherence of motion across the upper body, hands, torso, and lower body. The questionnaire interface used in the study is illustrated in Fig. 4.

Prior to evaluation, participants are provided with instructions explaining the four criteria and the five-point scoring scheme. We collect the ratings from all participants and compute the MOS for each method and criterion. To assess the consistency of subjective judgments, inter-rater reliability is measured using Krippendorff's alpha, yielding $\alpha = 0.78$. We further conduct pairwise Wilcoxon signed-rank tests between IntentTalk and the competing methods, with statistical significance evaluated at $p < 0.01$. The resulting subjective scores are reported in the main paper, where IntentTalk achieves the highest MOS across all four perceptual criteria.

REFERENCES

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